

A Phase–Riesz Refinement of the Large-Dimension Argument for the Lonely Runner Problem

Xiyu Hu

Abstract

For a set $V \subset \mathbb{N}$ of n distinct positive integer speeds, write

$$\text{ML}(V) = \max_{t \in \mathbb{T}} \min_{v \in V} \|tv\|_{\mathbb{T}}.$$

Bedert proved the uniform asymptotic lower bound

$$\text{ML}(V) \geq \frac{1}{2n} + \frac{1}{n^{5/3+o(1)}}.$$

We refine the large-additive-dimension branch of his argument and prove

$$\text{ML}(V) \geq \frac{1}{2n} + \frac{1}{n^{13/8+o(1)}}.$$

The new ingredient is a Riesz product with independent phases. The covering inequality produces a nonnegative function on phase space with very small mean. A level-two Fourier inequality then forces the quadratic signed-sum coefficients of the covering function to have small ℓ^2 -energy. Bedert’s arithmetic-progression intersection estimate converts this into an ℓ^1 -bound for the quadratic Riesz chaos with an additional factor equal to the covering excess. The remaining Riesz levels begin cubically. Balancing the resulting large-dimension estimate with Bedert’s short-model estimate yields the exponent $13/8$.

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1 Introduction

The Lonely Runner Conjecture is a classical problem in Diophantine approximation and combinatorics; we refer to the recent survey of Perarnau and Serra for its history and equivalent formulations [6]. For a finite set $V = \{v_1, \dots, v_n\} \subset \mathbb{N}$ of distinct positive integers, define

$$\text{ML}(V) = \max_{t \in \mathbb{T}} \min_{v \in V} \|tv\|_{\mathbb{T}}.$$

The conjecture asserts that $\text{ML}(V) \geq 1/(n+1)$. The elementary union bound gives $\text{ML}(V) \geq 1/(2n)$. Perarnau and Serra obtained a constant-order improvement at scale n^{-2} [5]; Tao obtained a logarithmically growing improvement [8]; and Bedert proved the polynomial estimate

$$\text{ML}(V) \geq \frac{1}{2n} + \frac{1}{n^{5/3+o(1)}}. \quad (1.1)$$

Our main result improves the exponent $5/3$.

Theorem 1.1. *There exist absolute constants $c, C > 0$ such that, for every sufficiently large n and every set $V \subset \mathbb{N}$ of n distinct positive integers,*

$$\text{ML}(V) \geq \frac{1}{2n} + \frac{c}{n^{13/8}(\log n)^C}. \quad (1.2)$$

Equivalently,

$$\text{ML}(V) \geq \frac{1}{2n} + \frac{1}{n^{13/8+o(1)}}.$$

Bedert's proof divides according to the two-additive dimension $d = \dim_2(V)$. His large-dimension branch uses a Riesz product and gives, up to powers of $\log n$,

$$\text{ML}(V) - \frac{1}{2n} \gg \frac{d^2}{n^3}, \quad (1.3)$$

whereas his small-dimension branch gives

$$\text{ML}(V) - \frac{1}{2n} \gg \frac{1}{nd} n^{-o(1)}. \quad (1.4)$$

The two estimates balance at $d = n^{2/3+o(1)}$, yielding (1.1) [1].

We refine the first branch. The quadratic layer of Bedert's Riesz product is normally treated by a raw triangle inequality. By adding independent phases to the product, the pointwise covering inequality gives a nonnegative function of very small mean on a high-dimensional torus. A level-two Fourier estimate makes the quadratic signed-sum energy small and inserts the covering excess into the quadratic error term. The untreated Riesz expansion then begins cubically.

At the level of exponents, write

$$\varepsilon = 2n \text{ML}(V) - 1.$$

The refined quadratic error is proportional to $\rho^2 \varepsilon n^{1/2+o(1)}$, while the remaining error begins at $\rho^3 n^{o(1)}$. Taking $\rho = n^{-1/4-o(1)}$ makes the cubic error negligible at scale $n^{-5/8}$. The large-dimension threshold becomes $d = n^{5/8+o(1)}$, and the small-dimension branch then gives a gap of order $n^{-13/8-o(1)}$.

Organisation

Section 2 fixes notation and states the precise inputs imported from Bedert. Section 3 constructs the phase Riesz product and identifies its level-two Fourier coefficients. Section 4 proves the level-two inequality. Section 5 converts the resulting ℓ^2 -bound into the ℓ^1 -bound needed in the zero-phase Riesz product. Section 6 carries out the modified large-dimension argument, and Section 7 balances it with the small-dimension branch. Section 8 explains the relation to the higher-moment and relation-lattice program.

2 Set-up and imported ingredients

We write $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, $e(x) = e^{2\pi i x}$, and

$$\widehat{f}(r) = \int_{\mathbb{T}} f(x) e(-rx) dx \quad (r \in \mathbb{Z}).$$

Let

$$\delta = \text{ML}(V), \quad I_\delta = [-\delta, \delta] \subset \mathbb{T}, \quad \varphi = \mathbf{1}_{I_\delta},$$

and define the covering multiplicity

$$\Phi(x) = \Phi_{V,\delta}(x) := \sum_{v \in V} \varphi(vx). \quad (2.1)$$

By the definition of δ , the sets $\{x : \|vx\|_{\mathbb{T}} \leq \delta\}$ cover $\mathbb{T}$, and hence

$$1 \leq \Phi(x) \leq n \quad (x \in \mathbb{T}). \quad (2.2)$$

Moreover,

$$\int_{\mathbb{T}} \Phi = 2n\delta = 1 + \varepsilon, \quad \varepsilon := 2n\delta - 1 \geq 0. \quad (2.3)$$

Thus

$$\delta - \frac{1}{2n} = \frac{\varepsilon}{2n}. \quad (2.4)$$

We may freely assume $\delta \leq 0.9/n$, since otherwise the desired conclusion is much weaker than the available constant-factor gap.

The Fourier expansion of φ is

$$\varphi(x) = 2\delta + \sum_{j \neq 0} \lambda(j) e(jx), \quad \lambda(j) = \frac{\sin(2\pi j \delta)}{\pi j}. \quad (2.5)$$

Consequently,

$$\widehat{\Phi}(r) = \sum_{\substack{v \in V \\ v|r}} \lambda(r/v) \quad (r \neq 0), \quad (2.6)$$

and, since $\delta \asymp 1/n$ in the relevant range,

$$|\lambda(j)| \ll \min\left(\frac{1}{n}, \frac{1}{|j|}\right). \quad (2.7)$$

Definition 2.1 (Two-dissociation). A finite set $D \subset \mathbb{Z}$ is *two-dissociated* if

$$\sum_{d \in D} c_d d = 0, \quad c_d \in \{-2, -1, 0, 1, 2\},$$

implies $c_d = 0$ for every $d \in D$. The maximum cardinality of such a subset of V is denoted $\dim_2(V)$.

Fix a two-dissociated set $D \subset V$ with $|D| = d := \dim_2(V)$. The following three statements are imported from Bedert's argument [1].

Proposition 2.2 (Fejér selection [1, Lemma 4.2 and (11)]). *There is an integer $j_0 \in [100n]$ such that, for the dilate*

$$D' := j_0 D,$$

one has

$$\sum_{m \in D'} \widehat{\Phi}(m) \geq c \frac{d}{n} \quad (2.8)$$

for an absolute constant $c > 0$.

For $0 \leq k \leq d$, define the signed-sum layer

$$E_k(D') = \left\{ \sum_{m \in D'} \sigma_m m : \sigma_m \in \{-1, 0, 1\}, \sum_m |\sigma_m| = k \right\}. \quad (2.9)$$

Two-dissociation implies that the signed representation of every element of $\bigcup_k E_k(D')$ is unique, and the sets $E_k(D')$ are pairwise disjoint.

Proposition 2.3 (Arithmetic-progression intersection [1, Lemma 4.3]). *There is an absolute constant C such that for every arithmetic progression $P \subset \mathbb{Z}$,*

$$|E_k(D') \cap P| \leq (C \log(2 + |P|))^k. \quad (2.10)$$

The slightly weaker version with an additional factor $\log |D'|$ is also sufficient throughout.

Proposition 2.4 (Small-dimension branch [1, Propositions 2.4 and 2.5]). *There are absolute constants $c, C > 0$ such that*

$$\text{ML}(V) - \frac{1}{2n} \geq \frac{c}{nd(\log n)^C} - O(n^{-100}). \quad (2.11)$$

The rest of the paper refines the complementary large-dimension branch.

3 The phase Riesz product

For a phase vector $\theta = (\theta_m)_{m \in D'} \in \mathbb{T}^{D'}$, define

$$R_\theta(x) = \prod_{m \in D'} (1 + \cos 2\pi(mx + \theta_m)). \quad (3.1)$$

The plus sign is deliberate: this product is not the one used for the final contradiction. Its purpose is to convert the pointwise covering inequality into positivity on phase space.

Lemma 3.1 (Normalisation and positivity). *For every $\theta \in \mathbb{T}^{D'}$,*

$$R_\theta(x) \geq 0, \quad \int_{\mathbb{T}} R_\theta(x) dx = 1. \quad (3.2)$$

Proof. Nonnegativity is immediate because every factor lies in $[0, 2]$. Expanding $1 + \cos 2\pi y = 1 + \frac{1}{2}e(y) + \frac{1}{2}e(-y)$ gives

$$R_\theta(x) = \sum_{\sigma \in \{-1, 0, 1\}^{D'}} 2^{-|\text{supp } \sigma|} e\left(x \sum_m \sigma_m m\right) e\left(\sum_m \sigma_m \theta_m\right). \quad (3.3)$$

A term contributes to the integral in x only if $\sum_m \sigma_m m = 0$. Since D' is two-dissociated, this forces $\sigma = 0$, proving that the constant term is one. $\square$

Define

$$Z(\theta) = \int_{\mathbb{T}} \Phi(x) R_\theta(x) dx, \quad Y(\theta) = Z(\theta) - 1. \quad (3.4)$$

Lemma 3.2 (A nonnegative small-mean phase function). *The function $Y : \mathbb{T}^{D'} \rightarrow \mathbb{R}$ satisfies*

$$0 \leq Y(\theta) \leq n - 1, \quad \mathbb{E}_\theta Y(\theta) = \varepsilon. \quad (3.5)$$

Proof. By [Equations \(2.2\)](#) and [\(3.2\)](#), $Z(\theta)$ is the average of a function taking values in $[1, n]$ against a probability density. Thus $1 \leq Z \leq n$. Averaging [Equation \(3.1\)](#) over all phases leaves only the constant term, so

$$\mathbb{E}_\theta R_\theta(x) = 1.$$

Therefore

$$\mathbb{E}_\theta Y(\theta) = \int_{\mathbb{T}} \Phi(x) dx - 1 = \varepsilon. \quad \square$$

The key point is the exact relation between the phase Fourier coefficients of Y and the signed-sum Fourier coefficients of Φ .

Lemma 3.3 (Fourier identification). *For every nonzero $\sigma \in \{-1, 0, 1\}^{D'}$,*

$$\hat{Y}(\sigma) = 2^{-|\text{supp } \sigma|} \hat{\Phi} \left(- \sum_{m \in D'} \sigma_m m \right). \quad (3.6)$$

Moreover, the map

$$\sigma \mapsto \sum_m \sigma_m m$$

is injective on $\{-1, 0, 1\}^{D'}$.

Proof. Insert [Equation \(3.3\)](#) into [Equation \(3.4\)](#) and integrate in x . This gives

$$Z(\theta) = \sum_{\sigma} 2^{-|\text{supp } \sigma|} \hat{\Phi} \left(- \sum_m \sigma_m m \right) e \left(\sum_m \sigma_m \theta_m \right).$$

The zero coefficient is $\hat{\Phi}(0) = 1 + \varepsilon$, and subtracting one only changes that zero coefficient. For injectivity, if two vectors σ, τ give the same sum, then

$$\sum_m (\sigma_m - \tau_m) m = 0$$

with coefficients in $\{-2, -1, 0, 1, 2\}$; two-dissociation gives $\sigma = \tau$. $\square$

4 A level-two inequality on phase space

We need a standard Fourier-level estimate for a bounded function of small mean. The ambient group is a product torus, but only the characters with coordinate frequencies in $\{-1, 0, 1\}$ are used.

Let $\mathcal{H}_2$ denote the span of the characters

$$e(\sigma \cdot \theta), \quad \sigma \in \{-1, 0, 1\}^d, \quad |\text{supp } \sigma| = 2.$$

Write P_2 for the orthogonal projection onto $\mathcal{H}_2$.

Lemma 4.1 (Degree-two Steinhaus hypercontractivity). *For every $q \geq 2$ and every $g \in \mathcal{H}_2$,*

$$\|g\|_q \leq Cq \|g\|_2 \quad (4.1)$$

with an absolute constant C .

Remark 4.2. This is the order-two case of the Bonami inequality for Steinhaus chaos; see, for example, [2, 4]. For completeness, a direct random-partition proof from the complex Khintchine inequality is given in [Section A](#). Only the linear dependence $O(q)$ is used here.

Lemma 4.3 (Level-two inequality). *Let $f : \mathbb{T}^d \rightarrow [0, 1]$ be measurable and let $\int f = \mu$. Then*

$$\sum_{\substack{\sigma \in \{-1, 0, 1\}^d \\ |\text{supp } \sigma| = 2}} |\hat{f}(\sigma)|^2 \leq C\mu^2 \log^2 \frac{e}{\mu}. \quad (4.2)$$

The right-hand side is interpreted as zero when $\mu = 0$.

Proof. If $\mu = 0$, then $f = 0$ almost everywhere. Suppose $\mu > 0$. By duality,

$$\|P_2 f\|_2 = \sup_{\substack{g \in \mathcal{H}_2 \\ \|g\|_2 = 1}} |\langle f, g \rangle|. \quad (4.3)$$

Choose

$$q = \max \left(2, \left\lceil \log \frac{e}{\mu} \right\rceil \right), \quad p = \frac{q}{q-1}.$$

Since $0 \leq f \leq 1$,

$$\|f\|_p^p \leq \int f = \mu, \quad \|f\|_p \leq \mu^{1/p} = \mu^{1-1/q} = \mu \exp \left(\frac{\log(1/\mu)}{q} \right) \leq e\mu.$$

By Hölder and [Lemma 4.1](#), every g in [Equation \(4.3\)](#) satisfies

$$|\langle f, g \rangle| \leq \|f\|_p \|g\|_q \leq C\mu q \leq C\mu \log \frac{e}{\mu}.$$

Squaring the resulting estimate for $\|P_2 f\|_2$ proves the claim by Parseval. $\square$

Apply [Lemma 4.3](#) to

$$f(\theta) = \frac{Y(\theta)}{n}, \quad \mu = \frac{\varepsilon}{n}.$$

By [Lemma 3.2](#), $0 \leq f \leq 1$. By [Lemma 3.3](#), every level-two coefficient of f is $1/(4n)$ times a unique Fourier coefficient of Φ on $E_2(D')$. We obtain the central energy estimate.

Proposition 4.4 (Quadratic energy bound). *One has*

$$\sum_{\ell \in E_2(D')} |\hat{\Phi}(\ell)|^2 \leq C\varepsilon^2 \log^2 \frac{en}{\varepsilon}. \quad (4.4)$$

When $\varepsilon = 0$, the left-hand side vanishes.

5 From quadratic energy to an ℓ^1 bound

The zero-phase Riesz product requires the sum of absolute values, not merely the energy. The conversion uses [Proposition 2.3](#) to show that only $n^{1+o(1)}$ relevant quadratic frequencies can have a small Fourier multiplier index.

Fix

$$J = n^3 \tag{5.1}$$

and define

$$\mathcal{S} = E_2(D') \cap \bigcup_{v \in V} v[-J, J]. \tag{5.2}$$

Lemma 5.1 (Low-multiplier support). *One has*

$$|\mathcal{S}| \leq Cn(\log n)^2. \tag{5.3}$$

Proof. For each $v \in V$, the set $v[-J, J]$ is an arithmetic progression of size $2J+1$. By [Proposition 2.3](#) with $k = 2$,

$$|E_2(D') \cap v[-J, J]| \leq C \log^2(2J+3) \leq C \log^2 n.$$

Summing over v proves the claim. $\square$

Lemma 5.2 (Low-multiplier contribution). *One has*

$$\sum_{\ell \in \mathcal{S}} |\widehat{\Phi}(\ell)| \leq C\epsilon n^{1/2} \log n \log \frac{en}{\epsilon}. \tag{5.4}$$

Proof. Cauchy–Schwarz, [Lemma 5.1](#), and [Proposition 4.4](#) give

$$\sum_{\ell \in \mathcal{S}} |\widehat{\Phi}(\ell)| \leq |\mathcal{S}|^{1/2} \left(\sum_{\ell \in E_2(D')} |\widehat{\Phi}(\ell)|^2 \right)^{1/2},$$

which is exactly the stated bound. $\square$

Lemma 5.3 (High-multiplier tail). *One has*

$$\sum_{\ell \in E_2(D') \setminus \mathcal{S}} |\widehat{\Phi}(\ell)| \leq C \frac{(\log n)^2}{n^2}. \tag{5.5}$$

Proof. By [Equations \(2.6\)](#) and [\(2.7\)](#), and by the definition of $\mathcal{S}$,

$$\sum_{\ell \in E_2(D') \setminus \mathcal{S}} |\widehat{\Phi}(\ell)| \leq \sum_{v \in V} \sum_{\substack{j \in \mathbb{Z} \\ |j| > J \\ jv \in E_2(D')}} \frac{1}{|j|}.$$

Fix v . For $r \geq 0$, place the positive and negative integers with $2^r J < |j| \leq 2^{r+1} J$ into two arithmetic progressions. By [Proposition 2.3](#), the number of such j with $jv \in E_2(D')$ is at most

$$C \log^2(2 + 2^{r+1} J).$$

Hence the contribution for this v is at most

$$\frac{C}{J} \sum_{r \geq 0} \frac{\log^2(2 + 2^{r+1} J)}{2^r} \leq C \frac{\log^2(2J)}{J}.$$

Summing over n values of v and using $J = n^3$ proves the claim. $\square$

Combining the two pieces gives the estimate that replaces the raw quadratic bound in Bedert's proof.

Proposition 5.4 (Quadratic ℓ^1 -bound). *Let*

$$Q_2 := \sum_{\ell \in E_2(D')} |\widehat{\Phi}(\ell)|.$$

Then

$$Q_2 \leq C\varepsilon n^{1/2} \log n \log \frac{en}{\varepsilon} + C \frac{(\log n)^2}{n^2}. \quad (5.6)$$

6 The modified large-dimension branch

We now return to Bedert's zero-phase Riesz product. To avoid confusing the Riesz parameter with a prime, we denote it by ρ :

$$\mathcal{R}_\rho(x) = \prod_{m \in D'} (1 - \rho \cos 2\pi m x), \quad 0 \leq \rho \leq 1. \quad (6.1)$$

As in [Lemma 3.1](#), two-dissociation implies

$$\mathcal{R}_\rho \geq 0, \quad \int_{\mathbb{T}} \mathcal{R}_\rho = 1.$$

Expanding by signed-sum level gives

$$\widehat{\mathcal{R}_\rho}(\ell) = \left(-\frac{\rho}{2}\right)^k \quad \text{for } \ell \in E_k(D'), \quad (6.2)$$

and zero otherwise.

Testing the pointwise inequality $\Phi \geq 1$ against this probability density and using [Proposition 2.2](#) yields

$$\begin{aligned} 1 &\leq \int_{\mathbb{T}} \Phi(x) \mathcal{R}_\rho(x) dx \\ &= 1 + \varepsilon - \rho \sum_{m \in D'} \widehat{\Phi}(m) + \frac{\rho^2}{4} \sum_{\ell \in E_2(D')} \widehat{\Phi}(\ell) + T_{\geq 3} \\ &\leq 1 + \varepsilon - c\rho \frac{d}{n} + \frac{\rho^2}{4} Q_2 + |T_{\geq 3}|. \end{aligned} \quad (6.3)$$

It remains to bound the higher levels beginning with $k = 3$.

Proposition 6.1 (Riesz levels three and above). *There is an absolute constant C such that, whenever*

$$C\rho^{1/3} \log n \leq \frac{1}{2}, \quad (6.4)$$

one has

$$|T_{\geq 3}| \leq C\rho^3 (\log n)^9 + n^2 \rho^{10 \log n}. \quad (6.5)$$

Proof. Using Equation (2.6) and the triangle inequality as in Bedert's large-dimension argument [1, Section 4],

$$|T_{\geq 3}| \leq 2n \max_{v \in V} \sum_{k=3}^d \rho^k \sum_{j \geq 1} |\lambda(j)| \mathbf{1}_{E_k(D')}(jv). \quad (6.6)$$

Let $K = \lceil 10 \log n \rceil$. For $k \geq K$, the disjointness of the layers and $|\lambda(j)| \leq 1/j$ give

$$\sum_{k=K}^d \rho^k \sum_{j \geq 1} |\lambda(j)| \mathbf{1}_{E_k}(jv) \leq \rho^K \sum_{\substack{j \geq 1 \\ jv \in \bigcup_{k \geq K} E_k}} \frac{1}{j}.$$

The union has at most $3^d \leq 3^n$ elements, so the last harmonic sum is $O(n)$. After the factor $2n$ in Equation (6.6), this contributes $O(n^2 \rho^{10 \log n})$.

Now take $3 \leq k < K$. By Equation (2.7), split the j -sum into $[1, n]$ and the blocks $[n^r, n^{r+1})$, $r \geq 1$. Applying Proposition 2.3 to each block gives

$$\begin{aligned} \sum_{j \geq 1} |\lambda(j)| \mathbf{1}_{E_k}(jv) &\leq \frac{1}{n} |E_k \cap v[1, n]| + \sum_{r \geq 1} \frac{1}{n^r} |E_k \cap v[n^r, n^{r+1})| \\ &\leq \frac{1}{n} (C \log n)^k \left(1 + \sum_{r \geq 1} \frac{(r+1)^k}{n^{r-1}} \right) \\ &\leq \frac{1}{n} (C \log n)^{2k} (1 + k^k) \\ &\leq \frac{1}{n} (C \log n)^{3k}, \end{aligned}$$

where the final step uses $k \leq 10 \log n$. Inserting this into Equation (6.6) gives

$$|T_{\geq 3}| \leq n^2 \rho^{10 \log n} + C \sum_{k=3}^{\infty} (C \rho^{1/3} \log n)^{3k}.$$

Under Equation (6.4), the series is geometric and is bounded by its first term, which is $O(\rho^3 \log^9 n)$. $\square$

Combining Equations (5.6), (6.3) and (6.5) yields

$$\begin{aligned} 0 \leq \varepsilon - c \rho \frac{d}{n} + C \rho^2 \varepsilon n^{1/2} \log n \log \frac{en}{\varepsilon} + C \rho^2 \frac{(\log n)^2}{n^2} \\ + C \rho^3 (\log n)^9 + n^2 \rho^{10 \log n}. \end{aligned} \quad (6.7)$$

Proposition 6.2 (Refined large-dimension estimate). *There exist absolute constants $c, C_1, C_2 > 0$ such that, for all sufficiently large n ,*

$$d \geq n^{5/8} (\log n)^{C_1} \quad (6.8)$$

implies

$$\text{ML}(V) - \frac{1}{2n} \geq \frac{c}{n^{13/8} (\log n)^{C_2}}. \quad (6.9)$$

Proof. Write $L = \log n$. Suppose, toward a contradiction, that

$$\varepsilon \leq c_0 n^{-5/8} L^{-M} \quad (6.10)$$

for a large fixed M and a sufficiently small c_0 . Choose

$$\rho = c_1 n^{-1/4} L^{-B}, \quad (6.11)$$

where B is a large fixed constant and $c_1 > 0$ is small. Then Equation (6.4) holds for large n .

The favourable term in Equation (6.7) satisfies, by Equation (6.8),

$$\rho \frac{d}{n} \geq c_1 n^{-5/8} L^{C_1 - B}. \quad (6.12)$$

The function $x \mapsto x \log(en/x)$ is increasing for all $0 < x \leq 1$. Therefore Equation (6.10) implies

$$\varepsilon \log \frac{en}{\varepsilon} \leq C c_0 n^{-5/8} L^{1-M}. \quad (6.13)$$

Since $\rho^2 n^{1/2} = c_1^2 L^{-2B}$, the quadratic term containing ε is at most

$$C c_0 c_1^2 n^{-5/8} L^{2-M-2B}.$$

The purely high-multiplier quadratic tail is $O(n^{-5/2} L^{O(1)})$. The cubic error is

$$C \rho^3 L^9 = C c_1^3 n^{-3/4} L^{9-3B} = o\left(n^{-5/8} L^{C_1 - B}\right),$$

and $n^2 \rho^{10 \log n}$ is superpolynomially smaller.

Choose the constants in the order B , then $C_1 > B$, then M , and finally choose c_0, c_1 sufficiently small. The sum of all positive terms in Equation (6.7), including the initial ε , is then strictly smaller than the negative term in Equation (6.12), a contradiction. Thus $\varepsilon \geq c n^{-5/8} L^{-C_2}$ for some C_2 , and Equation (2.4) gives the result. $\square$

7 Balancing the two branches

We now combine Proposition 6.2 with Bedert's small-dimension estimate.

Proof of Theorem 1.1. Let C_1 be the threshold exponent from Proposition 6.2. If

$$d \geq n^{5/8} (\log n)^{C_1},$$

then Proposition 6.2 applies. Otherwise, Proposition 2.4 gives

$$\text{ML}(V) - \frac{1}{2n} \geq \frac{c}{n d (\log n)^{C_0}} - O(n^{-100}) \geq \frac{c}{n^{13/8} (\log n)^{C_0 + C_1}} - O(n^{-100}).$$

Increasing the final logarithmic exponent absorbs the negligible error. $\square$

Remark 7.1 (Where the gain occurs). Bedert's unrefined quadratic error has size

$$O(\rho^2 (\log n)^{O(1)}),$$

which forces $\rho \lesssim d/n$ and yields d^2/n^3 . The phase estimate replaces it by

$$O\left(\rho^2 \varepsilon n^{1/2} (\log n)^{O(1)}\right).$$

Under a contradiction hypothesis at the $13/8$ scale, this term is absorbable for the fixed choice $\rho = n^{-1/4 - o(1)}$, so the first uncontrolled chaos is cubic.

8 Relation to the higher-moment and relation-lattice program

The argument above treats the first analytic obstruction—the quadratic Riesz layer. It was motivated by a broader higher-moment and relation-lattice program. This section records the rigorous part of that program and explains where it may enter next.

Let

$$X(x) = \Phi(x) - 1 \geq 0, \quad M_r = \int_{\mathbb{T}} X(x)^r dx.$$

Under a near-extremal covering hypothesis, one has $M_1 = \varepsilon$ and a uniform positive lower bound for M_2 . The moments are log-convex:

$$M_r^2 \leq M_{r-1} M_{r+1}. \quad (8.1)$$

Therefore the ratios M_{r+1}/M_r are nondecreasing, and

$$M_r \geq M_2 \left(\frac{M_2}{M_1} \right)^{r-2} \gg \varepsilon^{-(r-2)}. \quad (8.2)$$

This is the correct proof of high-moment growth. An induction that substitutes a lower bound for M_{r-1} into the denominator of $M_{r+1} \geq M_r^2/M_{r-1}$ does not justify the same conclusion.

Expanding $\int \Phi^r$ turns Equation (8.2) into many large intersections of Bohr sets. Through Bohr–GAP duality, those intersections correspond to boxes containing many points of relation lattices. The appropriate structural parameters are not the ordinary rational rank of scalar speeds—that rank is always one—but rather:

- the effective or proper rank at coefficient scale $O(n)$;
- the two-dissociative or Freiman dimension;
- the successive-minimum profile of the projected relation lattice;
- the compatibility of short relation labels on overlapping tuples.

A large box count controls the product of successive minima, but does not by itself force one specific “many short, one long” profile. One must first divide into dyadic profiles and separate balanced and unbalanced cases.

The local-to-global step also requires more than ordinary hypergraph regularity. Each good tuple carries an integer relation label. A useful theorem must exploit agreement of these labels on overlaps. In particular, dense but independently labelled packets can realise the large moment numerology without producing one global recurrence.

8.1 The next quantitative target

The present proof controls the quadratic layer by a support bound

$$|\mathcal{S}| \leq n^{1+o(1)}.$$

Suppose a structural packet/agreement theorem improved the effective quadratic support to

$$M \leq \left(\frac{n}{d} \right)^{2+o(1)}. \quad (8.3)$$

The same phase-energy argument would then permit a larger Riesz parameter, leaving a genuinely cubic optimisation. The resulting large-dimension branch would have the schematic form

$$\text{ML}(V) - \frac{1}{2n} \gtrsim \frac{d^{3/2}}{n^{5/2+o(1)}}.$$

Balancing with $1/(nd)$ gives $d = n^{3/5+o(1)}$ and the exponent $8/5$.

More generally, if all chaos layers below order s could be controlled and the first untreated error were $O(\rho^s n^{o(1)})$, the expected balanced exponent would be

$$\frac{3s-1}{2s-1}. \quad (8.4)$$

The first values are shown in [Table 1](#).

First untreated chaos	Expected gap exponent	Interpretation
2	5/3	Bedert's original balance
phase control of level 2	13/8	present argument
3	8/5	quadratic packet/agreement target
4	11/7	one further structural layer
$s \rightarrow \infty$	3/2	natural limit of the hierarchy

Table 1: The proposed hierarchy of improvements.

Thus the higher-moment/lattice program remains central beyond the first improvement. The phase Riesz product removes the first analytic obstruction and exposes the more structural packet problem underneath.

A A direct proof of degree-two Steinhaus hypercontractivity

We give a self-contained reduction of [Lemma 4.1](#) to the standard complex Khintchine inequality. Let $z_j = e(\theta_j)$, and for $\alpha \in \{+1, -1\}$ write

$$z_j^{[\alpha]} = \begin{cases} z_j, & \alpha = +1, \\ \bar{z}_j, & \alpha = -1. \end{cases}$$

Every $g \in \mathcal{H}_2$ has a unique decomposition

$$g = \sum_{\alpha, \beta \in \{+1, -1\}} g_{\alpha, \beta}, \quad g_{\alpha, \beta} = \sum_{1 \leq i < j \leq d} a_{ij}^{\alpha, \beta} z_i^{[\alpha]} z_j^{[\beta]}. \quad (\text{A.1})$$

The four pieces are orthogonal in L^2 . It is therefore enough, up to an absolute factor, to treat one fixed sign pair. Write

$$Q(z) = \sum_{i < j} a_{ij} z_i^{[\alpha]} z_j^{[\beta]}.$$

Choose a random subset $S \subset [d]$ by placing each index in S independently with probability $1/2$, and define the cross-part

$$Q_S(z) = \sum_{\substack{i < j \\ i \in S, j \notin S}} a_{ij} z_i^{[\alpha]} z_j^{[\beta]}.$$

Every ordered term survives with probability $1/4$, so

$$Q = 4\mathbb{E}_S Q_S.$$

Minkowski's inequality gives

$$\|Q\|_q \leq 4\mathbb{E}_S \|Q_S\|_q. \quad (\text{A.2})$$

Fix S . Conditional on the variables $(z_i)_{i \in S}$, write

$$Q_S = \sum_{j \notin S} b_j(z_S) z_j^{[\beta]}, \quad b_j(z_S) = \sum_{\substack{i \in S \\ i < j}} a_{ij} z_i^{[\alpha]}.$$

The complex Khintchine inequality (see, e.g., [3]) states that, for $q \geq 2$,

$$\left\| \sum_j c_j z_j \right\|_q \leq C \sqrt{q} \left(\sum_j |c_j|^2 \right)^{1/2}.$$

It is unchanged when some z_j are replaced by $\bar{z}_j$. Applying it first in the variables outside S gives

$$\|Q_S\|_q \leq C \sqrt{q} \left\| \left(\sum_{j \notin S} |b_j|^2 \right)^{1/2} \right\|_{L^q(z_S)}.$$

Since $q/2 \geq 1$, Minkowski's inequality in $L^{q/2}$ yields

$$\left\| \left(\sum_j |b_j|^2 \right)^{1/2} \right\|_q \leq \left(\sum_j \|b_j\|_q^2 \right)^{1/2}.$$

A second application of complex Khintchine, now in the variables indexed by S , gives

$$\|b_j\|_q \leq C \sqrt{q} \left(\sum_{\substack{i \in S \\ i < j}} |a_{ij}|^2 \right)^{1/2}.$$

Consequently,

$$\|Q_S\|_q \leq C q \left(\sum_{i < j} |a_{ij}|^2 \right)^{1/2} = C q \|Q\|_2.$$

Together with Equation (A.2), this proves the required estimate for each sign type. Summing the four pieces in Equation (A.1) proves

$$\|g\|_q \leq C q \|g\|_2,$$

which is Lemma 4.1.

Acknowledgements

The phase refinement arose from the author's higher-moment and relation-lattice approach to the problem. The proof builds on Bedert's additive-dimension decomposition and arithmetic-progression intersection estimate.

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